

Sergio Monteiro

Bachelor's & Master's in Aerospace Engineering | Embry-Riddle Aeronautical University, USA

Aerospace Engineer | Avionics Systems & GNC

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PROFILE

Holds bachelor's and master's degrees in Aerospace Engineering from Embry-Riddle Aeronautical University, USA. Aerospace engineer based in Warton within the Eurofighter Avionics Joint Team, with broad aircraft avionics knowledge and experience in system-level definition and integration across communications, automatic ground collision avoidance (AGCAS), and armament equipment including laser designating pods and air-to-air missiles. Background spans guidance, navigation and control (GNC), Airbus configuration management, and multidisciplinary collaboration throughout the product lifecycle.

EDUCATION

Master's in Aerospace Engineering

Embry-Riddle Aeronautical University | Daytona Beach, Florida, USA | May 2023

Concentration: Aerospace Guidance, Navigation, Dynamics & Controls

Bachelor's Degree in Aerospace Engineering

Embry-Riddle Aeronautical University | Prescott, Arizona, USA | May 2021

Concentration: Astronautics

TECHNICAL EXPERTISE

Systems: Avionics integration; requirements definition and traceability; interfaces and protocols; hazard analysis; lifecycle planning; design maturity reviews; configuration management.

GNC: Formation control; collision avoidance; task allocation; Kalman filtering; spacecraft attitude dynamics; orbital mechanics; linear, nonlinear, optimal and adaptive control.

Tools: DOORS, MATLAB, Simulink, GMAT, CATIA V5, ANSYS, NASTRAN, VHDL, LabVIEW.

PROFESSIONAL EXPERIENCE

Avionics Systems Engineer - Eurofighter Avionics Joint Team (AJT)

Airbus Defence and Space | Warton, UK | Sep 2025 - Present

- Contribute to system-level definitions across Eurofighter avionics, including the Automatic Ground Collision Avoidance System (AGCAS), laser designating pods and air-to-air missiles, addressing functional requirements and interfaces within the wider aircraft avionics architecture.
- Integrate Eurofighter communications systems as part of the Fighters Communications System Integration Team (TADYC-1), covering data links including MIDS, VMF, CESMO and JREAP.
- Work with Eurofighter consortium partners to define avionics system and subsystem designs, contractual interoperability requirements, and Level 2 functional requirements, interfaces and protocols.
- Develop detailed system-element design information consistent with architecture models and allocated requirements; maintain requirements and traceability in DOORS and contribute to hazard analysis.
- Identify development lifecycle activities to meet regulatory and programme assurance requirements; define checkpoints, reviews and maturity expectations aligned with programme phases and gates.
- Collaborate with Engineering, Manufacturing Engineering and industrial and functional design disciplines from early design through serial production and in-service support, accounting for industrial planning infrastructure, constraints and limitations.

PROFESSIONAL EXPERIENCE - CONTINUED

GNC Engineer

SENER Aerospace & Defense | Seville, Spain | Jan 2024 - Jul 2025

- Held technical responsibility for Swarm Formation Control and Resource Management work packages on SWAT-SHOAL, a European defence project for cooperative unmanned underwater vehicles (UUVs) operating with degraded communications.
- Designed and implemented MATLAB/Simulink formation control algorithms using leader-follower structures, waypoint tracking, virtual-point navigation and trajectory adjustments.
- Developed collision avoidance using cost-function minimisation and potential fields, with autonomous formation reconfiguration to navigate obstacles.
- Implemented task allocation and reassignment based on vehicle proximity, battery levels and task complexity, alongside mission monitoring and safe return logic.
- Applied Kalman filtering and time-delay compensation to support navigation accuracy and formation synchronisation in GNSS-denied environments.

Aerospace Configuration Management

Sogecclair Aerospace - Airbus Operations | Seville, Spain | Jul 2023 - Dec 2023

- Managed engineering changes across A320, A330, A340 and A350 programmes, from change initiation through implementation and modification (MOD) closure.
- Coordinated TRS and SubTRS activities with the design office, reviewing technical solutions and documentation for wing, fuel, autoflight and surveillance systems.
- Organised technical and decision meetings, prepared agendas and recorded minutes in accordance with established change processes.
- Coordinated weekly Change Request Reviews (CRRs) to assess readiness for Kick-off, Maturity and Decision Boards.

SELECTED ACADEMIC PROJECTS

Embry-Riddle Aeronautical University

- **Satellite attitude control:** Developed a MATLAB/Simulink controller for ground-station pointing. Applied optimal control to limit control torque under disturbances and a bank of Kalman filters to reduce simulated sensor noise.
- **Adaptive satellite vibration control:** Modelled nonlinear cyclic vibrations in satellite telemetry components and designed an adaptive Simulink controller to mitigate vibration.
- **Near-Earth Object mission:** Planned a rendezvous mission for a prospective asteroid-mining application. Used MATLAB to minimise transfer delta-V and GMAT to simulate transfer, rendezvous and return trajectories.
- **GEPETO rover:** Contributed to rover design and construction, taking responsibility for telemetry and data acquisition. Developed MATLAB code to interpret analogue and digital LabJack sensor data; designed the structure, solar panels, wheels and suspension in CATIA V5.
- **Lunar dust mitigation:** Collaborated on the design, construction and testing of an active system to reduce lunar dust accumulation on spacesuits.

AWARDS & LEADERSHIP

Academic recognition: Best AE Undergraduate Student, Outstanding AE Graduate and Top Student Scholar Award (2021); Top 10 NAIA Daktronics Scholar Athletes (2019 - 2023); Dean's List (2017 - 2023).

Leadership & teaching: Captain, Embry-Riddle Men's Soccer Team (2017 - 2023); Public Relations, Student-Athlete Advisory Committee (2019 - 2023); Graduate Teaching Assistant (2021 - 2023).

Honour societies: Phi Kappa Phi (since 2021); Tau Beta Pi (since 2019); Sigma Gamma Tau (since 2018).

Scholarships: Presidential, academic and soccer scholarships (2017 - 2023).

LANGUAGES

English, Spanish, French and Portuguese.